



Nikhil Suresh
Data Scientist & Machine Learning Expert

Data Scientist (13+ Years' Experience)
(Power BI, Python, AWS, Big Data, Databricks, NLP, Gen AI)

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JOB SKILLS

- Certified Data Scientist with a **Master's in Data Science** and over **6+ years** of experience across various domains (**Insurance**, banking, finance, healthcare, transport, quick commerce).
- **NLP Research & Development:**
 - Worked extensively with Large Language Models (LLMs) such as **OpenAI's** ChatGPT and **Huggingface's** models, utilizing them for diverse tasks, from conversational agents to content generation and prompt designing using Zero-Shot Prompt & Few-Shot Prompt.
 - Used Agentic Frameworks like **OpenAI SKD**, **CrewAI**, **Langgraph** and **Autogen**.
 - Integrated **QLoRA-based fine-tuning** pipelines with **Hugging Face's transformers** and **PEFT** libraries for scalable and efficient training.
 - Created **HLD [High Level Design]** and **LLD [Low Level Design]** for the entire AI project.
 - Constructed custom **Goggle GenerativeAI** , **Gemini Pro Vision** model that labels pictures files , using Streamlit web application.
 - Created **Retrieval augmentation Generation [RAG]** system on **Streamlit** using OpenAI, Pinecone Vector Database, and LangChain summarizing PDF, MS Word, Text document which reduced the decision-making process by 80%.
 - Built NLP-based solutions for risk mitigation and fraud detection (**topic modeling (LDA)**, **text summarization**, named entity recognition (**NER**) - tech stack: **BERT**, gensim, nltk, spacy)
- **Machine Learning Deployment Framework:**
 - Experienced with data visualization tools like **Tableau** and **PowerBI**.
 - Created Web Services with **Flask** Framework to Deploy Machine learning models in ****AWS EC2**** instance as IAAS and also on **Palantir Foundry** and **Heroku** as PAAS (using **Flask** Framework) and used **** Kubernetes **** for orchestration.
 - Created a E2E package/pipeline, viz extraction of **Topics/Telemetry** data from **Sensors IoT-Devices**, exported the same to **Confluent Kafka**, migrated the data from **Kafka** to **MongoDB** performed transformations, saved the artifacts/output into ****AWS S3**** bucket, and created a frontend using ****FAST API****, built a docker image of the source code, saved the docker image in ****AWS ECR**** and completed with orchestration with **GIT Actions**.
 - Developed containerized (Docker Image) deployments of ML frameworks (e.g. Spark, H2O, distributed TensorFlow, **Airflow**) to a **Kubernetes** cluster.
 - Architected solutions with **Python (NumPy, Pandas, SciPy)**, Scikit-Learn, **LSTM**, Neural Networks, Transformers, Arima, Prophet, **TensorFlow**, skforecast, XGBoost, Random Forest
 - Achieved a **15%** increase in accuracy in spam email classification using Bayesian networks implemented with Python libraries such as **NumPy, Pandas, SciPy, Scikit-Learn, XGBoost**.
 - Expert in Cloud Technology Management – Containerization, Spark, **Linux**, **CI/CD** in **Azure**
 - Reduced false positives in **fraud detection** by **20%** using probabilistic graphical models alongside **Scikit-Learn, Random Forest, LSTM, Deep Neural Networks**.
 - Implemented a **recurrent neural network (RNN)** architecture resulting in a **15%** improvement in time-series prediction accuracy, enhancing TS forecasting capabilities.

Technical Forte

Cloud	Microsoft Azure, AWS, GCP
Big Data Platform	Azure Synapse, Azure ML Studio, Spark MLlib, Hadoop
Dev Platform	PyCharm, Jupyter Notebook, AWS Sagemaker , VS Code
Visualization Package	Tableau, PowerBI, Power BI, Matplotlib, Seaborn, Plotly
CI/CD	Docker, Kubernetes, Azure Pipelines, GITlab, Flask, HTML,GIT actions,
Programming Languages	Python (OOP), R, Scala, Apache Spark, MATLAB, PySpark, Shell Scripting, Linux, Bash Scripting
Database	SQL, NoSQL, Apache Cassandra, MongoDB, HQL, Jira, Kanban Board
LLM / Deep Learning Framework	LangChain, OpenAI, Google Gemini Vision Pro, Hugging Face, Transformers, TensorFlow, Keras, PyTorch, Theano, CNTK, CrewAI

EXPERIENCE

May 2025 – Present

Senior GenAI Engineer

Encora Innovation Labs, Bangalore, India [Consultant to Eli-Lilly]

Gen Solutions & Innovations:

- Developed a multi-agent virtual assistant to aid clinicians with medical summarization , diagnostic insights and treatment recommendations, using **openAI SDK** and **CrewAI** as Agentic Framework., deployed to production on **Vercel**.
- Developed a **multimodal RAG** system that generates spoken responses from the LLM, leveraging **OpenAI's speech-to-text** capabilities for audio processing.
- Built **NextJS in React** with Typescript, with Pagere Router(client-side), **FastAPI** backend, **Tailwind CSS** & deployed to **Vercel**, used for user authentication.
- Integrated advanced LLM's including NVIDIA NIM LLM's like NVIDIA Palmyra and third party tools like Tavily and Wikipedia to enhance assistant capabilities
- Implemented a **ReACT-based agent workflow** leveraging OpenAI as the core reasoning engine for dynamic decision-making.
- Led design and deployment of a **custom PyTorch/Transformers handwriting OCR** for names; trained on **SageMaker** with **EventBridge-scheduled** retraining and **Lambda-driven** deployment; established **data versioning** and **confidence-based fusion** with Textract for robust production inference.
- Automated a **closed-loop training pipeline: user corrections → curated S3 dataset (images + CSV labels) → weekly SageMaker training (30 epochs) → auto-deploy** to real-time endpoint; delivered **scalable batching** (≤45 images) and **parallel inference** for throughput.
- Utilized **Langchain, LangGraph** MCP Servers and clients to orchestrate multi-agent communication and enable tool calling functionality
- Deployed the application on AWS EC2 and exposed it via a RESTful API using FASTapi.
- Containerized the full solution with Docker to enable scalable and portable deployment

Data Scientist Expert

Merck KGaA [Sigma Aldrich], Bangalore, India

NLP Solutions & Innovations:

- Worked with **Azure OpenAI Studio** and **Azure AI Studio**, using closed OpenAI models and open source models like **FLAN-T5** , **LLama-2**, **Minstral v2** etc.
- Optimized data analytics pipeline to improve real-time data processing capabilities for a retail client. The project utilized a robust tech stack including AWS for cloud services, **PostgreSQL** as our **relational database** system, and **PySpark** for processing large datasets efficiently.
- Optimized ****LLaMA-Instruct 70B**** an open-source **Hugging Face** model for Merck's Healthcare domain-specific tasks by leveraging **Quantized Low-Rank Adaptation (QLoRA)**, reducing memory footprint while maintaining model performance.
- Evaluated the quality of machine-generated translations from **German to English** for Merck's Healthcare department using the **SacreBLEU** & **ROGUE** metric, leveraging the ****Opus-Medical-German-to-English**** **Hugging Face** model.
- Created Question-Answer **RAG** system using **LangChain**, **FASSIS** vector DB and **OpenAI** for Talent Acquisition Team and deployed it in **** AWS EC2 **** instance using FASTapi to create frontend web portal, used index.html & used **** Kubernetes **** for orchestration, which reduced the decision-making process by **80%**.
- Developed & deployed **BERT-based text classification** model for **sentiment analysis**, intent detection & **topic categorization**, improving **accuracy** by **11%**.
- Experienced with data visualization tools like **Tableau** and **PowerBI**.

Leadership/Team Management:

- Leading team of 3 data scientists, review the team's algorithms, code and reduced code complexity in an effective manner, increasing team efficiency by 26%.
- Mentored junior team members on **QLoRA** and advanced NLP techniques, fostering a culture of innovation and knowledge sharing.

Sept 2023– Nov 2024

Oct 2021 – Sept 2023

● **Data Science Architect**

HP, Bangalore, India

NLP Solutions & Innovations:

- Developed and fine-tuned **time series forecasting models** using open-source pretrained models like **IBM Granite TTM-R2** and **Lag-LLaMA** for accurate trend prediction.
- Integrated **BERT-based classifiers** into real-time inference pipelines, reducing classification latency by **29%** using techniques like model **quantization**.
- Streamlined end-to-end pipeline development cycle of **LLM-based** applications from ideation, prototyping, testing evaluation to production deployment and monitoring using managed service like **Azure Prompt Flow & AWS Sagemaker**.
- Optimized **learning rate, epochs, and optimizer algorithms (Adam, AdamW, RMSprop)** to enhance model convergence and stability.
- Exploratory data analysis (Python), developed predictive machine models (time series) & extracted actionable insights.
- Reduced deployment time for ML models from weeks to minutes by containerizing the applications using **Docker** and **Kubernetes**.

Leadership/Team Management:

- Encouraged a collaborative code review culture, fostering peer feedback and discussions to enhance problem-solving, improve code quality, and drive continuous learning within the team

Nov 2020 – Sept 2021

● **Data Scientist [Freelancer, Contract: Remote]**

Lock & Stock, Dubai, UAE

- Managed over 1 TB of data in **Microsoft Azure SQL Database**, achieving a 40% improvement in performance by optimizing indexes and queries, significantly enhancing report generation speeds.
- Developed and optimized speech-to-text pipelines using **OpenAI's Whisper-Turbo** model for high-accuracy transcription from Arabic to English.
- Optimized existing machine learning models to reduce the runtime there by achieving reduction in production cost by **15%**.
- Improved speech recognition accuracy by **25%** using **recurrent neural networks (RNNs)** with long short-term memory (**LSTM**) cell
- Reduced error rates in image recognition tasks by **40%** through the adoption of **convolutional neural networks (CNNs)**.

Leadership/Team Management:

- Overseeing model evaluation and validation, ensuring performance assessment using key metrics (accuracy, precision, recall, **F1-score**) while enforcing proper techniques like cross-validation to maintain robustness and alignment with business objectives.

June 2017 – Oct 2020

● **Machine Learning Engineer**

Galaxy, Gurgaon, India

- Reduced the runtime of existing Machine Learning models by optimizing various parameters from **3 days to 7.2 hours on AWS Sagemaker**.
- Developing tools for analyzing and processing large datasets, extracting insights, and preparing data for AI models.
- Achieved a **30%** reduction in prediction latency through kernelized methods for stock price forecasting.
- Delivered customer-centric solutions leveraging machine learning and deep learning methodologies, resulting in a **40%** increase in customer satisfaction scores and a **30%** decrease in customer churn rates

Achievements

- Improved customer churn prediction accuracy by 25% by employing **Support Vector Machines (SVMs)** with Gaussian kernel functions.

ACADEMIA

- Masters in Data Science & AI
Bachelor in Business Management
- Simplilearn, Bangalore (India)
Bangalore University (India)

LANGUAGE

- English – Fluent
German-(Beginner)

PERSONAL SKILLS

- Self-motivated performer, **strong organizational skills, creative thinker, adaptable** and **technologically competent** to achieve any project milestone
- Empathic listener and **persuasive** speaker, Influencing, leading, and delegating abilities
- Confident, **articulate**, & **professional** speaking abilities (and experience)